

Researchers Use Chemical Examinations and Human Smell to Study Ancient Remains

A recent study about the smell of Egyptian mummies reached some surprising findings. The ancient bodies did not smell bad, the researchers found. Instead, they say, the remains mostly smelled good.

Cecilia Bembibre is director of research at University College London's Institute for Sustainable Heritage. "In films and books, terrible things happen to those who smell mummified bodies," she said. "We were surprised at the pleasantness of them."

The researchers reported the leading descriptions of the smells as "woody," "spicy" and "sweet." They also reported a floral or flowery smell. That smell could be from pine and juniper resins used in mummification; a process designed to protect the body from decay.

In this undated photo provided by Emma Paolin, a view of a selection of the mummified bodies in the exhibition area of the Egyptian museum in Cairo. (Emma Paolin via AP)

The study appeared recently in the Journal of the American Chemical Society. It used both chemical examinations and several human smellers to study nine mummies. The mummies, some around 5,000 years old, had been housed at the Egyptian Museum in Cairo.

Bembibre, one of the report's writers, said the researchers wanted to study the smell of mummies because it has long been a subject of interest for the public and researchers alike.

She added that even fiction writers have written pages of work on the subject — for good reason.

Scent, or smell, was an important consideration in the mummification process. This process used oils and plant-based materials to protect the body and its spirit for the afterlife. Mummification was largely used for powerful people such as pharaohs and other leaders.

Pleasant smells were linked with purity and gods. Bad smells were signs of corruption and decay.

Researchers did not want to directly measure the mummies, because doing so might be damaging. Instead, researchers were able to take measurements that did not involve touching the mummies. Researchers from UCL and the University of Ljubljana in Slovenia were able to measure smells, pesticides, and other effects due to mold, bacteria or microorganisms.

Using technical instruments to measure air molecules released from sarcophagi was very important, said Matija Strlič, a chemistry professor at the University of Ljubljana.

"It tells us potentially what social class a mummy was from and therefore reveals a lot of information about the mummified body...." he said. "We believe that this approach is potentially of huge interest to other types of museum collections."

Barbara Huber of the Max Planck Institute of Geoanthropology in Germany was not involved in the study. Huber said the findings provide important data on compounds that could preserve or damage mummified remains. The information could be used to better protect the ancient bodies for future generations.

Huber said that over thousands of years, differing conditions have changed the scents of the mummies in a major way.

Huber wrote a study two years ago that examined a jar that had contained mummified organs of a woman. The goal was to identify the material used to preserve the organs and what that would show about ancient trade paths.

Researchers of the current study hope to do something similar. They want to use their findings to develop "smellscape" to recreate the scents they discovered. They also want to change the experience for future museumgoers.

Bembibre said museums generally ask visitors to experience everything with their eyes. She added that seeing mummies through "a glass case reduces the experience because we don't get to smell them."

Smelling the scents of mummification would improve the museum visitors' experiences, she suggested, as smell is one of the ways that people understand the world.

I'm John Russell.

Brian Melley reported on this story for the Associated Press. John Russell adapted it for VOA Learning English.

Vocabulary

mummy - n. ร่างกายโบราณที่ได้รับการเก็บรักษาไว้สำหรับการฝังศพ

resin - n. สารอินทรีย์ธรรมชาติที่มักจะโปร่งใสหรือโปร่งแสงและมีสีเหลืองถึงน้ำตาล และมักทำจากวัสดุพืช; สารเหล่านี้อะลายได้ในตัวทำละลายอินทรีย์แต่ไม่ละลายในน้ำ ไม่นำไฟฟ้า และใช้สำหรับเคลือบและป้องกันพื้นผิว

decay - n. การสลายตัวหรือทำให้สลายตัว

fiction - n. สิ่งที่ถูกสร้างขึ้นโดยจินตนาการ

museum - n. สถานที่ที่อุทิศให้กับการดูแล ศึกษา และจัดแสดงวัตถุที่มีความน่าสนใจ ความสำคัญ หรือคุณค่า

sarcophagus - n. ภาชนะโบราณที่ใช้บรรจุศพมนุษย์

potentially - adv. เป็นไปได้

reveal - v. แสดงให้เห็นอย่างชัดเจน

approach - n. วิธีการจัดการกับสิ่งใดสิ่งหนึ่ง

preserve - v. ทำให้คงอยู่ในสภาพดี

Reading Comprehension Quiz

1. What was the surprising finding of the recent study on Egyptian mummies?

- A. The mummies had no smell at all.
- B. The mummies smelled mostly pleasant.
- C. The mummies smelled terrible, as expected.
- D. The mummies smelled strongly of preservatives.

2. According to Cecilia Bembibre, what is a common portrayal of mummified bodies in films and books?

- A. They are shown as objects of great historical value.
- B. They are often depicted with a pleasant aroma.
- C. They are associated with terrible events for those who smell them.
- D. They are usually ignored for their sensory aspects.

3. Which of the following was NOT among the leading descriptions of the mummy smells reported by researchers?

- A. Woody
- B. Spicy
- C. Fruity
- D. Sweet

4. What was a possible source for the floral or flowery smell detected on the mummies?

- A. The ancient textiles used for wrapping.
- B. The natural scent of the human remains.
- C. Pine and juniper resins used in mummification.
- D. The environment of the museum where they were stored.

5. Where was the recent study about mummy smells published?

- A. University College London's Institute for Sustainable Heritage journal.
- B. The Journal of the American Chemical Society.
- C. The Associated Press news report.
- D. The Egyptian Museum's official publication.

6. How many mummies were examined in the study mentioned in the article?

- A. Two
- B. Five
- C. Nine
- D. Fifteen

7. Why did researchers, including Bembibre, decide to study the smell of mummies?

- A. To challenge common misconceptions from films and books.
- B. Because it was a topic of long-standing interest to both the public and researchers.
- C. To identify new methods of preservation for ancient remains.
- D. To understand the cultural significance of ancient Egyptian burial rituals.

8. What does the article suggest about the involvement of fiction writers with the subject of mummy smells?

- A. They have contributed inaccurate portrayals of the smells.
- B. They have shown considerable interest in the topic.
- C. They have criticized the scientific methods used in such studies.
- D. They have provided historical evidence about mummification.

9. For whom was the mummification process primarily used?

- A. All members of ancient Egyptian society.
- B. Slaves and common laborers.
- C. Powerful individuals like pharaohs and leaders.
- D. Priests and religious figures exclusively.

10. In ancient Egyptian beliefs, what did pleasant smells signify?

- A. Wealth and social status.
- B. Purity and association with gods.
- C. The effectiveness of the preservation process.
- D. Readiness for the afterlife journey.

11. Why did the researchers avoid directly touching the mummies during their measurements?

- A. They lacked the necessary protective equipment.
- B. It was considered disrespectful to the ancient remains.
- C. Direct contact might cause damage to the mummies.
- D. The mummies were too fragile to be moved.

12. According to Matija Strlič, what was a crucial method for measuring smells from the mummies?

- A. Direct chemical analysis of mummy tissues.
- B. Using human smellers to identify scents.
- C. Measuring air molecules released from sarcophagi with technical instruments.
- D. Analyzing the materials used in mummification.

13. What potential benefit does Matija Strlič suggest this measurement approach offers?

- A. It can determine the exact age of the mummy.
- B. It can reveal the social class of the mummified individual.
- C. It helps identify the cause of death for the individual.
- D. It can recreate the exact smells for public display.

14. According to Barbara Huber, what is a significant application of the study's findings regarding compounds?

- A. To create more authentic museum exhibits.
- B. To develop new dating techniques for ancient artifacts.
- C. To better preserve ancient bodies for future generations.
- D. To understand the ancient Egyptian diet.

15. What was the primary goal of Barbara Huber's earlier study on a jar containing mummified organs?

- A. To determine the social status of the woman whose organs were mummified.
- B. To identify the materials used for preservation and learn about ancient trade paths.
- C. To recreate the original smell of the mummified organs.
- D. To compare the preservation methods with modern techniques.